VANISHING OF THE HOMOLOGY OF SYMMETRIC POWERS BELOW THE SYMMETRIC DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the notation and assumptions fixed in a cross-chapter example and lemma; both are declared as navigation pointers and the proof works with that notation.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The cosimplicial resolution computation of homology on the punctual topos. Cohomology on Sites, Example sites-cohomology-example-category-to-point and Lemma sites-cohomology-lemma-compute-by-cosimplicial-resolution, used at proof lines 2283-2287. Fixes the setting of the statement.
- The Eilenberg-Zilber type formula for lower shriek of a tensor product. Cohomology on Sites, Lemma sites-cohomology-lemma-eilenberg-zilber, used at proof lines 2315-2321. Turns the E_1 terms into derived tensor products of derived lower shrieks.
- The Koszul complex resolving the symmetric powers. used at proof lines 2298-2307. Produces the exact complex and hence the spectral sequence.
- Flatness of F and the vanishing H_0(C, F) = 0. used at proof lines 2288-2289 and 2325-2329. Give the base cases and the vanishing for wedge powers.

Source reference. Original section 08PF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11023–11023. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08Q9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11052–11052. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11055–11055. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Notation and assumptions as in Cohomology on Sites, Example 08PF. Assume $\mathcal{C}$ has a cosimplicial object as in Cohomology on Sites, Lemma 08Q9. Let $\mathcal{F}$ be a flat $\underline{B}$-module such that $H_0(\mathcal{C}, \mathcal{F}) = 0$. Then $H_l(\mathcal{C}, Sym^k_{\underline{B}}(\mathcal{F})) = 0$ for $l < k$.*

Proof. We drop the subscript $\underline{B}$ from tensor products, wedge powers, and symmetric powers. We will prove the lemma by induction on k. The cases $k = 0, 1$ follow from the assumptions. If $k > 1$ consider the exact complex

$$\ldots \to \wedge^2 \mathcal{F} \otimes \mathrm{Sym}^{k-2} \mathcal{F} \to \mathcal{F} \otimes \mathrm{Sym}^{k-1} \mathcal{F} \to \mathrm{Sym}^k \mathcal{F} \to 0$$

with differentials as in the Koszul complex. If we think of this as a resolution of $\mathrm{Sym}^k \mathcal{F}$, then this gives a first quadrant spectral sequence

$$E_1^{p,q} = H_p(\mathcal{C}, \wedge^{q+1} \mathcal{F} \otimes \mathrm{Sym}^{k-q-1} \mathcal{F}) \Rightarrow H_{p+q}(\mathcal{C}, \mathrm{Sym}^k(\mathcal{F}))$$

By Cohomology on Sites, Lemma 08QC we have

$$L\pi_!(\wedge^{q+1} \mathcal{F} \otimes \mathrm{Sym}^{k-q-1} \mathcal{F}) = L\pi_!(\wedge^{q+1} \mathcal{F}) \otimes^{\mathbf{L}}_B L\pi_!(\mathrm{Sym}^{k-q-1} \mathcal{F}))$$

It follows (from the construction of derived tensor products) that the induction hypothesis combined with the vanishing of $H_0(\mathcal{C}, \wedge^{q+1}(\mathcal{F})) = 0$ will prove what we want. This is true because $\wedge^{q+1}(\mathcal{F})$ is a quotient of $\mathcal{F}^{\otimes q+1}$ and $H_0(\mathcal{C}, \mathcal{F}^{\otimes q+1})$ is a quotient of $H_0(\mathcal{C}, \mathcal{F})^{\otimes q+1}$ which is zero. □